

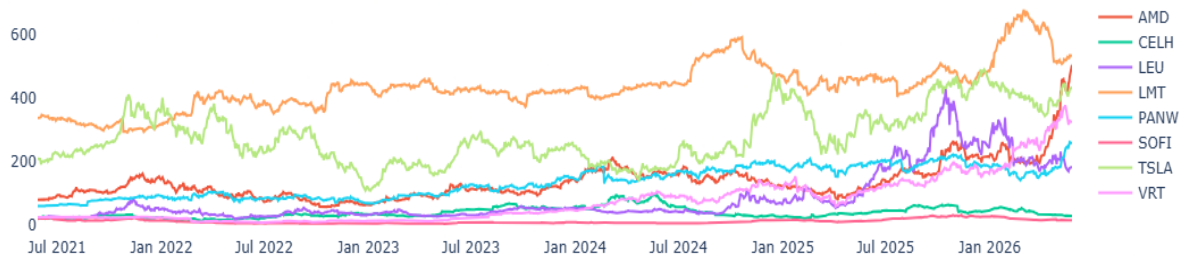
Portfolio Optimisation Using Python.

Key Objective:- Monte Carlo Simulations & Sharpe Ratio Maximization

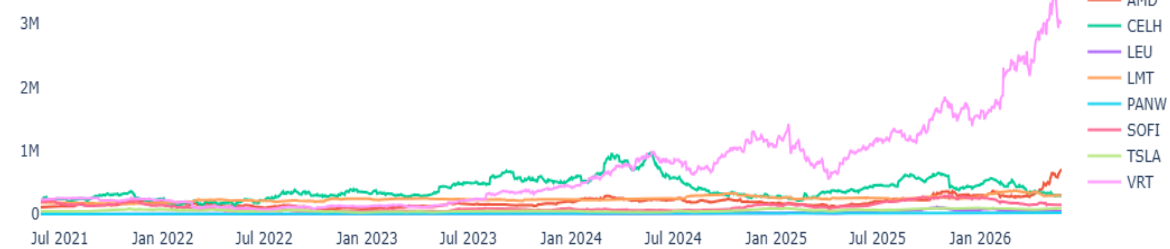
Key Takeaways:

- Risk-adjusted optimization significantly improved returns
- Diversification reduced portfolio volatility
- USD exposure outperformed INR purchasing power over 5 years
- Monte Carlo simulations helped identify efficient allocations

Adjusted Closing Prices [\$]



Portfolio positions [\$]



Total Portfolio Value [\$]

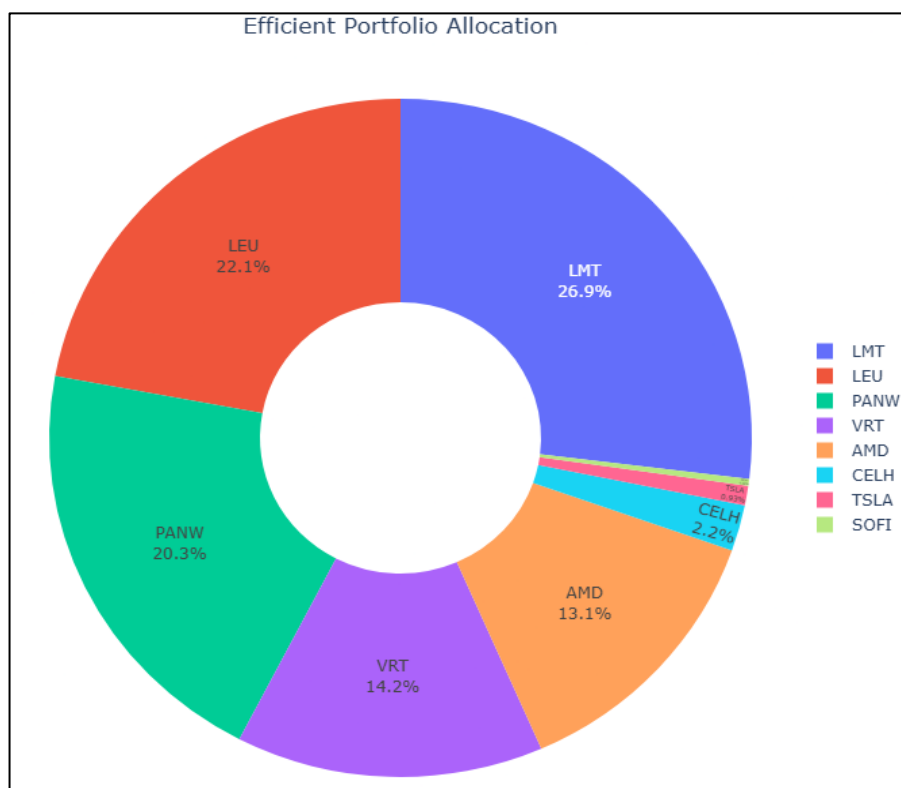


Best Portfolio Metrics Based on 1000 Monte Carlo Simulation Runs:

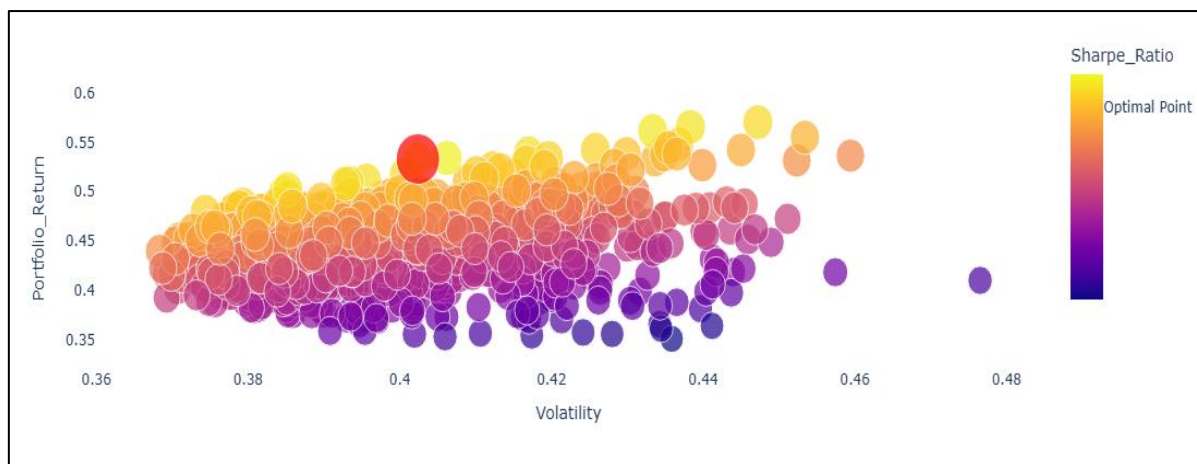
- Portfolio Expected Annual Return ----->	46.31%
- Portfolio Standard Deviation (Volatility) ----->	33.21%
- Sharpe Ratio ----->	1.28
- Final Value ----->	\$5853034.92
- Return on Investment ----->	485.30%

Stock Ideal_Weights

Ticker	%
AMD	13.05
CELH	2.20
LEU	22.07
LMT	26.93
PANW	20.34
SOFI	0.32
TSLA	0.93
VRT	14.15



Plotting The Efficient Frontier



Inefficient Portfolio vs Efficient Portfolio

